

Po-Yen Chen

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Profile

A machine learning engineer and M.Sc. in Simulation Sciences with an engineering-first mindset for turning research-grade code into production-ready ML systems. 3+ years building data and ML pipelines in Python, with hands-on MLOps across Docker, Infrastructure as Code (Terraform), workflow orchestration (Apache Airflow), and CI/CD. I enjoy being the bridge between scientists and scalable, maintainable software, with a strong bias toward clean code and modular architecture.

Skills

- **MLOps & DevOps:** Docker (containerized workloads), Terraform (Infrastructure as Code), Apache Airflow (workflow orchestration), CI/CD pipelines (GitHub Actions), Git, Linux — IaC and container skills transfer directly to the Azure / Ansible / Kubernetes ecosystem
- **ML lifecycle & tracking:** MLflow, DVC, Kubeflow (experiment tracking, data versioning, model lifecycle)
- **Cloud & data:** AWS (S3, EC2); SQL for large datasets
- **Python (expert):** clean, modular, design-pattern-driven code; Pandas, NumPy, scikit-learn, PyTorch, Matplotlib; also C and MATLAB
- **Machine learning:** data & ML model pipelines, physics-informed neural networks (PINNs) and operator learning (surrogate models), object detection (YOLO), LLM / RAG applications

Work Experience

AI Research Engineer | Industrial Technology Research Institute (ITRI), Taiwan | December 2025 – Present

- **AI4Sciences:** Accelerated molecular dynamics (particularly dissipative particle dynamics) and CFD simulations with physics-informed neural networks (PINNs) and operator learning as surrogate models for microfluidic design and particle-size distribution (PSD) prediction.
- **Agentic RAG:** Built and tuned an end-to-end agentic RAG pipeline with automated (“autoresearch”) tuning, and explored an LLM-driven knowledge base (llm-wiki) to streamline research workflows.

Engineer of Instrumentation & Control System | CTCTI, Taipei, Taiwan | April 2019 – March 2020

- Utilized VBA to automate tasks and enhance work efficiency; organized and designed instrumentation & control systems projects and layouts through effective cross-departmental collaboration.

Projects

Vehicle Recommendation System | Porsche Digital Campus Challenge 2024 | October 2024 – January 2025

- Customized a text embedding model using both Retrieval-Augmented Generation (RAG) and fine-tuning approaches to deliver insightful refined customer data. Developed a chatbot aimed at helping potential customers interested in Porsche vehicles learn more about specific Porsche products.

Improving Customer Experience with Generative AI | Hackathon EESTech Challenge Aachen | May 2024

- Web-scraped customer queries from the service website, designed the application integrating the LLM model, and performed prompt engineering to enhance the quality of online customer service.

A Machine Learning Playground in Quantum Mechanical Simulation | SiSc Laboratory 2022 | November 2022 – February 2023

- Developed an automated workflow through AiiDA for materials property prediction by implementing kernel ridge regression with structural descriptors (Coulomb matrix, SOAP) on 1730 topological insulator structures, establishing structure-property relationships for efficient prediction of magnetic properties.

Certification

Data Engineering Specialization | DeepLearning.AI with AWS on Coursera | October 2024

Algorithm Specialization | Stanford Online on Coursera | July 2024

Education

Master of Science in Simulation Sciences | RWTH Aachen University, Aachen, Germany | October 2021 – March 2025

- Relevant Courses: Machine Learning, Deep Learning in Physics Research, Data Analysis and Visualization, Model-based Estimation Methods, Numerical Optimization, High Performance Computing, Parallel Computing
- Thesis: Determination of Particle Size Distributions through Real-Time Area, Mass, and Geometry Flow Characterization of Non-Ferrous Metals: A YOLO-Based Instance Segmentation Approach

Bachelor of Science in Chemical Engineering | National Cheng Kung University (NCKU), Tainan, Taiwan | September 2015 – June 2018

Languages

English: Fluent; **German:** Intermediate; **Mandarin:** Native

Extracurricular

Smart City for Climate Adaption | Hackathon NRW with Remondis GmbH | November 2023

Event Organizer | Taiwan Studentenverein Aachen e.V. | April 2023 – Present

Interests

MLOps, DataOps, Data analysis and visualization, Deep learning, Bayesian statistics, Numerical optimization, Computer vision, Polymer science, Circular economy